

Weiwu Yan

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Education

University of Chicago	September 2023 – June 2025
• Master of Arts in Computational Social Science	
University of California, San Diego	September 2019 – June 2023
• Bachelor of Science in Social Psychology	
• Bachelor of Arts in Sociology	

Projects

Gender & Dialogic Emotion	Spring 2025
• Performed computational analysis using RoBERTa, Parts-of-Speech Tagging, and TF-IDF to analyze gender differences in dialogic emotion in role-playing video games.	
Chat Application	Fall 2024
• Designed a chat application using Flask and Node Package Manager with features including user accounts, channels, and real-time messaging with text, emojis, and images.	
Academic Research Trends	Spring 2024
• Used Multinomial Logistic Regression and Latent Dirichlet Allocation on an arXiv dataset of 1.7 million scholarly articles using PySpark on Amazon Web Services to analyze trends in academic research fields.	
Urban Gentrification & Socioeconomic Status	Spring 2024
• Designed an agent-based model using Mesa to simulate urban gentrification dynamics, examining how citizens and immigrants with varying socioeconomic status interact and relocate within a city.	
Funding & Academic Output	Winter 2023
• Collected 4,959 funding records from the U.S. National Science Foundation Division of Behavioral and Cognitive Sciences and scraped metadata of 207,370 scholarly articles by 3,230 identified researchers from Google Scholar.	
• Performed K-Means Clustering using TF-IDF to analyze the relationship between seven identified fields and total and average funding.	
• Used a Structural Topic Model to analyze trends in the popularity of 12 identified topics.	
• Performed document embeddings using SciBERT to analyze the impact of funding on research diversity.	
Video Game Classification	Winter 2023
• Performed genre classification on a Steam dataset of 75,776 video games using a Random Forest Classifier and a Convolutional Neural Network, leveraging textual descriptions and cover art.	
Political Orientation & Prison Perception	Spring 2023
• Performed statistical analysis on 20 participant responses collected via a Google survey using Excel to examine the relationship between political orientation and perceptions of prison functions for violent versus property crimes.	
Eyewitness Identification & Confidence	Summer 2022
• Conducted a between-subjects experiment with 78 participants to examine the discriminability of simultaneous versus sequential lineups and assess confidence in eyewitness identification.	
Moral Judgment & Occupation	Fall 2020
• Performed statistical analysis on 285 participant responses collected via a Qualtrics survey using Jamovi to examine the relationship between a mother's occupation and moral judgments of mothers who leave their children unsupervised.	

Skills

Programming: Python, JavaScript & R

Artificial Intelligence: Machine Learning, Deep Learning & Natural Language Processing

Data Analysis: Pandas, NumPy, SQL, SciPy, dplyr, Excel & Jamovi

Data Visualization: Plotly, ggplot2, Tableau, Shiny, & Matplotlib

High-Performance Computing: Cloud Storage & Cloud Computing

Web Development: Flask, React & Node Package Manager

Survey: Google Survey & Qualtrics

Languages

Chinese: Native Speaker

English: Fluent Speaker